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Experiment No.: 6 **Title: Particle Swarm Optimization for Function Optimization**

1. Aim

To implement Particle Swarm Optimization (PSO) for a multimodal continuous optimization problem and study the effect of inertia weight and acceleration coefficients on convergence.

2. Objectives

1. Understand the working principle of Particle Swarm Optimization.
2. Apply PSO to a multimodal continuous optimization problem.
3. Study the effect of inertia weight w on exploration and exploitation.
4. Study the effects of cognitive coefficient c_1 and social coefficient c_2 on convergence.
5. Compare the convergence behaviour for different parameter settings.

3. Problem Definition

Consider the two-dimensional Rastrigin function

$$\boxed{f(x,y) = 20 + x^2 + y^2 - 10\cos(2\pi x) - 10\cos(2\pi y)}$$

with

$$-5.12 \leq x, y \leq 5.12.$$

The objective is to minimize $f(x,y)$. The global optimum is $(x^*, y^*) = (0,0)$, for which $f(x^*, y^*) = 0$.

The Rastrigin function is multimodal because its periodic cosine terms produce many local minima. Therefore, the optimization process must balance exploration and exploitation.

4. Particle Swarm Optimization

Particle Swarm Optimization is a population-based stochastic optimization method in which each particle represents a candidate solution.

For a two-dimensional problem, the position and velocity of particle i at iteration t are represented by $\mathbf{x}_i^t$ and $\mathbf{v}_i^t$, respectively. Each particle maintains its personal best position $\mathbf{p}_i$, while the swarm maintains the global best position $\mathbf{g}$.

The velocity is updated as

$$\boxed{\mathbf{v}_i^{t+1} = w\mathbf{v}_i^t + c_1 r_1(\mathbf{p}_i - \mathbf{x}_i^t) + c_2 r_2(\mathbf{g} - \mathbf{x}_i^t)}$$

and the position is updated using

$$\boxed{\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}.}$$

Here, w is the inertia weight, c_1 is the cognitive coefficient, c_2 is the social coefficient, and $r_1, r_2 \sim U(0,1)$ are random numbers.

The three components of the velocity represent:

$$\underbrace{w\mathbf{v}_i^t}_{\text{inertia}} + \underbrace{c_1 r_1(\mathbf{p}_i - \mathbf{x}_i^t)}_{\text{cognitive}} + \underbrace{c_2 r_2(\mathbf{g} - \mathbf{x}_i^t)}_{\text{social}}.$$

The inertia component preserves part of the previous movement, the cognitive component attracts the particle towards its own best position, and the social component attracts it towards the swarm's best position.

5. Particle Movement in PSO

Figure 1 illustrates the combined effect of inertia, personal best, and global best on the movement of a particle.

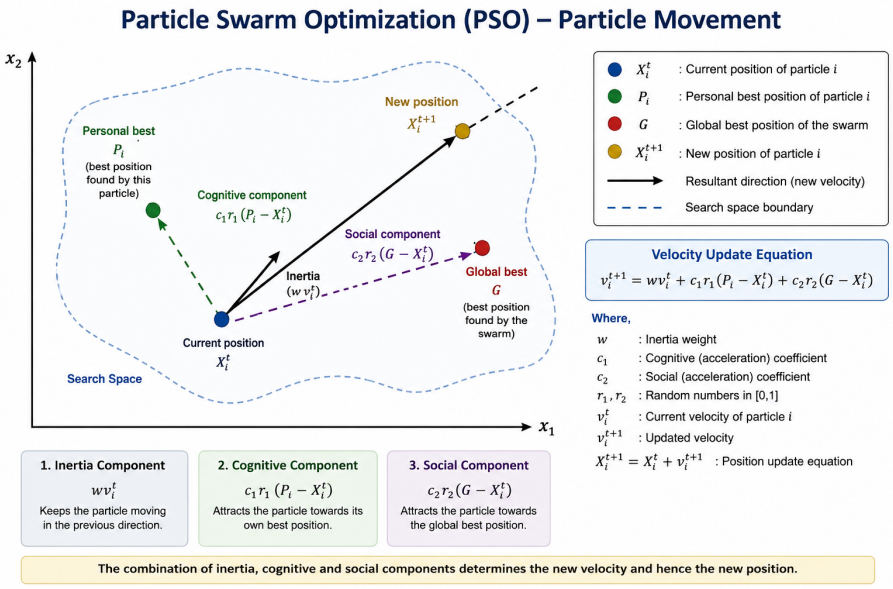


Figure 1: Particle movement under inertia, cognitive, and social components.

Thus, the new movement of a particle is determined jointly by its previous velocity, its own experience, and the experience of the swarm.

6. PSO Algorithm

Algorithm 1 Particle Swarm Optimization

Require: Number of particles N, maximum iterations T, search bounds, w, c_1, and c_2

Randomly initialize particle positions $\mathbf{x}_i$ and velocities $\mathbf{v}_i$ within the specified limits.
Evaluate the objective function for all particles.
Set $\mathbf{p}_i \leftarrow \mathbf{x}_i$ for every particle.
Determine the initial global best $\mathbf{g}$.
for $t = 1$ to T **do**
    for each particle i **do**
        Generate $r_1, r_2 \sim U(0, 1)$.
        Update velocity:

$$\mathbf{v}_i \leftarrow w\mathbf{v}_i + c_1 r_1(\mathbf{p}_i - \mathbf{x}_i) + c_2 r_2(\mathbf{g} - \mathbf{x}_i)$$

        Update position:

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i$$

        Apply boundary constraints.
        Evaluate $f(\mathbf{x}_i)$.
        if $f(\mathbf{x}_i) < f(\mathbf{p}_i)$ **then**
            $\mathbf{p}_i \leftarrow \mathbf{x}_i$
        end if
    end for
    Update $\mathbf{g}$ using the best personal-best position.
    Store the global-best fitness.
end for
return $\mathbf{g}$ and $f(\mathbf{g})$.

7. Numerical Example

Consider one particle with

$$\mathbf{x}_1 = (2, -1), \qquad \mathbf{v}_1 = (0.5, -0.2),$$

personal best

$$\mathbf{p}_1 = (1, -0.5),$$

and global best

$$\mathbf{g} = (0, 0).$$

Let

$$w = 0.5, \qquad c_1 = 1.5, \qquad c_2 = 2.0, \qquad r_1 = 0.4, \qquad r_2 = 0.6.$$

The inertia component is

$$w\mathbf{v}_1 = 0.5(0.5, -0.2) = (0.25, -0.10).$$

The cognitive component is

$$c_1 r_1(\mathbf{p}_1 - \mathbf{x}_1) = 1.5(0.4)(-1, 0.5) = (-0.6, 0.3).$$

The social component is

$$c_2 r_2(\mathbf{g} - \mathbf{x}_1) = 2(0.6)(-2, 1) = (-2.4, 1.2).$$

Therefore,

$$\mathbf{v}_1^{t+1} = (0.25, -0.10) + (-0.6, 0.3) + (-2.4, 1.2) = \boxed{(-2.75, 1.40)}.$$

The new position is

$$\mathbf{x}_1^{t+1} = (2, -1) + (-2.75, 1.40) = \boxed{(-0.75, 0.40)}.$$

For this position,

$$f(-0.75, 0.40) = 20 + (-0.75)^2 + (0.40)^2 - 10\cos(-1.5\pi) - 10\cos(0.8\pi)$$

and hence

$$\boxed{f(-0.75, 0.40) \approx 28.81}.$$

This fitness is compared with the particle's previous personal-best fitness. If the new fitness is smaller, the personal best is updated.

8. Experimental Study

Use the following common settings:

$$N = 30 \text{ particles}, \qquad T = 100 \text{ iterations},$$

with search range

$$[-5.12, 5.12]^2.$$

For each experiment, change one parameter while keeping the other two fixed.

8.1 Effect of Inertia Weight w

Set

$$c_1 = 2.0, \qquad c_2 = 2.0$$

and compare $w = 0.4,\ 0.7,\ 0.9$.

Table 1: Effect of inertia weight

Case	w	c_1	c_2	Best Fitness
Low inertia	0.4	2.0	2.0	
Medium inertia	0.7	2.0	2.0	
High inertia	0.9	2.0	2.0	

Observation: A larger inertia weight generally promotes exploration by allowing particles to retain more of their previous velocity. A smaller value generally promotes exploitation.

8.2 Effect of Cognitive Coefficient c_1

Set

$$w = 0.7, \qquad c_2 = 2.0$$

and compare $c_1 = 0.5,\ 1.5,\ 2.5$.

Table 2: Effect of cognitive coefficient c_1

Case	w	c_1	c_2	Best Fitness
Low c_1	0.7	0.5	2.0	
Medium c_1	0.7	1.5	2.0	
High c_1	0.7	2.5	2.0	

Observation: Increasing c_1 gives greater importance to the particle's own previous experience and personal best position.

8.3 Effect of Social Coefficient c_2

Set

$$w = 0.7, \qquad c_1 = 2.0$$

and compare $c_2 = 0.5,\ 1.5,\ 2.5$.

Table 3: Effect of social coefficient c_2

Case	w	c_1	c_2	Best Fitness
Low c_2	0.7	2.0	0.5	
Medium c_2	0.7	2.0	1.5	
High c_2	0.7	2.0	2.5	

Observation: Increasing c_2 increases the attraction towards the global best. A very large value may cause the swarm to converge prematurely towards the current global best.

9. Convergence Analysis

For each parameter setting, plot the global-best fitness against the number of iterations.

Iteration vs. Global Best Fitness

Compare the parameter settings using:

- final best fitness,
- convergence speed,
- stability of convergence,
- ability to avoid local minima.

Since PSO is stochastic, repeat each experiment for at least five independent runs and report the mean final fitness. The standard deviation may also be reported to measure the variability of the results.

10. Overall Observation Table

Table 4: Comparison of PSO parameter settings

Case	w	c_1	c_2	Best Fitness
Baseline	0.7	2.0	2.0	
Low w	0.4	2.0	2.0	
High w	0.9	2.0	2.0	
Low c_1	0.7	0.5	2.0	
High c_1	0.7	2.5	2.0	
Low c_2	0.7	2.0	0.5	
High c_2	0.7	2.0	2.5	

11. Expected Observations

- Increasing w generally increases exploration.
- Decreasing w generally promotes exploitation.
- Increasing c_1 strengthens individual learning.
- Increasing c_2 strengthens social learning.
- Excessively large parameter values may cause oscillatory or premature convergence.
- Good performance requires a suitable balance between exploration and exploitation.
- Different runs may produce different results because PSO uses random numbers.

12. Result

Particle Swarm Optimization was implemented for the multimodal Rastrigin function. The effects of inertia weight w, cognitive coefficient c_1, and social coefficient c_2 on convergence were studied experimentally.

The experiment demonstrates that PSO performance depends on the balance among particle momentum, individual learning, and social learning.

$$\boxed{\text{PSO movement} = \text{Inertia} + \text{Personal experience} + \text{Swarm experience}}$$